Branches of Linguistics:

- **Pragmatics:** meaning from conversational text
- **Semantics:** meaning of linguistic expressions
- **Syntax:** sentence structure
- **Morphology:** word structure
- **Phonology:** sound patterns, gesture systems
- **Phonetics:** signal, signed gesture, speech sounds

① Morphology

nouns: -s (plural), -ship
verbs: -ed, -ing, -en, -s (agreement), un- (reverse)
adjectives: -er, -est, un- (negation)
adverbs: -ly

② Syntactic distribution

↳ words in same position if appear in same position in string of words
↳ distinguishing environment: position where only members of one category can grammatically appear

e.g. The ___ exists
only nouns may fit

(Lexical categories)

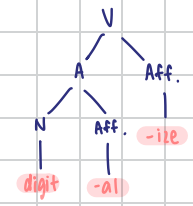
nouns:	det ___ verb	"the girl sang"
verbs:	will / to ___	"will arrive", "to go"
adjectives:	det ___ noun	"the happy dog"
adverbs:	verb ___ verb	"has quickly eaten"

Word Formation

① **Derivation:** attach affix to base to derive diff. category/meaning

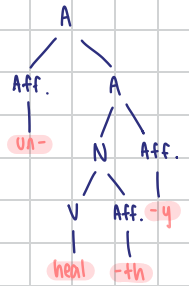
	V + [-able] → Adj	read → readable
	[ex-] + N → N	nun → ex-nun
homophonous	[un-] + A → A	happy → unhappy
homophonous	[un-] + V → V	wrap → unwrap
	[re-] + V → V	watch → rewatch
	A + [-ity] → N	act → activity

e.g. digitalize



digital, digitalize

unhealthy



health, healthy, unhealthy

② **Inflection:** required by position / function of word in sentence
↑ tense, agreement, number

e.g. Suri sing-s, Suri is sing-ing

Closure: inflection closes further morphological processes (derivation does not)

Case: nominative case (NOM.) on subjects
accusative case (ACC.) on objects

③ **Compounding:** endocentric: same semantic category with head (dog food)
exocentric: meaning is non-compositional (hot dog)

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Syntactic categories (Functional categories)

- **Determiners (Det):** a, an, the, some, few, many
- **Prepositions (Prep):** in, on, under, through, with
- **Preposition qualifiers (PQual):** just, right, straight [precede prepositions]
- **Degree words (Deg):** very, really
- **Negation (Neg):** not
- **Conjunction (Conj):** and, or, but
- **Tense (T):** [modal aux.] will, may, could, can, must
  [infinitival marker] to, -ed, -s
- **Complementizer (C):** that, if, for, whether

Root vs. affixes

roots: semantically complete in themselves
boyish, walkable, uncool

affixes: only meaningful in combination with other piece
boyish, walkable, uncool

Bound vs. free morphemes

free: stand alone (boy, walk, cool)
bound: attach to another morpheme (-ish, -able, un-)

Bound roots:	* scissor	* kempt
	scissors	unkempt
	scissor-kick	well-kempt

Types of affixes

- **suffix:** right of base — cat-s, teach-er
- **prefix:** left of base — un-happy, in-complete
- **infix:** inside base — b-um-uri
- **circumfix:** both sides of base — em-bold-en
- **zero-derivation:** null affix ∅ — staple (noun) + ∅ → staple (verb)
- **stem-change:** change in form — run (verb) + PAST → ran
  goose (noun) + PLURAL → geese
- **supra-segmental:** pitch, intensity, length — PERmit (noun) → perMIT (verb)
- **tone:** pitch (high, mid, low, falling, rising)

Allomorphy: morpheme with more than one form

cat - [s]	dog - [z]	size - [əz]
kick - [s]	friend - [z]	kiss - [əz]
pot - [s]	form - [z]	badge - [əz]

Constituents: grouping of words into larger units

Heads: every phrase contains a head which determines its category

Constituency tests

① **substitution test:** replace with pro-forms

NP → they/them, he/him, she/her, it, I/me, we

The girl in the red hat left her scarf on the subway.
She

VP → do so / did so

The girl in the red hat will buy a prom card.
do so

PP → there, here, then

The girl in the red hat left a scarf on the subway.
there

② **Movement test:** move a string around

↳ **clefting:** It was ___? that/who [sentence]

They will give a new fork to Guy.

It was to Guy [they will give a new fork].

↳ **pseudo-clefting:** what/who [sentence] is/was ___ who they will give fork to is Guy

↳ **topicalization:** ___, [sentence] To Guy, they will give a new fork.

③ **Fragment test:** fragment responses are constituents

What did the girl leave on the subway?

A red scarf

④ **Coordination test:** two strings coordinated (and, or, but) are both constituents

The kids ran down the road.

The kids ran down the road and around the bend.

We saw dogs and a group of cats

Phrase structure rules

NP → (Det) (AdjP+) N (PP+) (CP+) (The very silly very happy cat on the bed)

VP → (AdvP+) V (NP) (NP/CP) (AdvP+) (PP+) (AdvP+)

PP → (PQual) P (NP) (We ran right out the door)

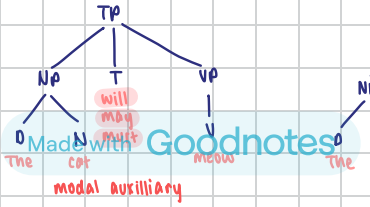
AdvP → (AdvP) Adv (The cat very quickly came home)

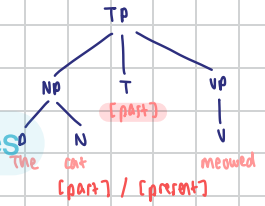
AdjP → (AdvP) Adj (PP) (I was very proud of my cat)

TP → NP/CP T VP

CP → C TP

VP: probably wrote the cat a book at noon quickly
AdvP V NP NP PP AdvP





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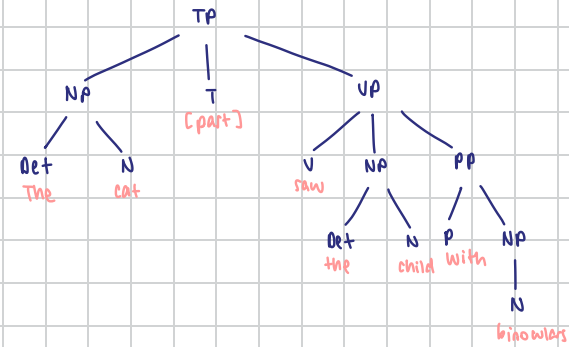
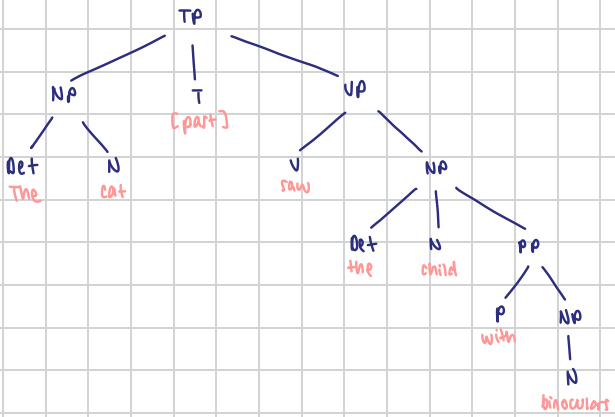
Structural ambiguity:

The cat saw the child with binoculars.





Dependants: phrase has head and dependants - complements / adjuncts

Complements

NP objects are complements of head V (selection by head)

Transitive: verb selects complement e.g. I like cats

Intransitive: verbs do not select complement e.g. * I arrived cats

Adjuncts

Typically correspond to modifiers (always optional, no selection by head)

The cat arrived at the party on Tuesday happily

Complementizer phrases

Declarative: statement / assertion The cat must know that she is sad.

Interrogative: question The cat will know if she is sad.

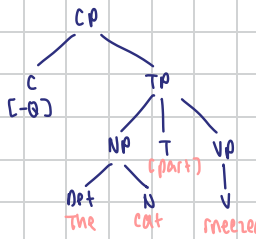
that [-Q]

if [+Q]

whether [+Q]

Add CP layer to all sentences, even if it is not embedded.

If there is no overt complementizer, mark as [+Q] or [-Q].



Clauses

Every clause has a main verb

=> count # of main verbs for # of clauses

Dependent vs. Independent Clauses

Independent: stand alone — The cat meowed.

Dependent: cannot stand alone — *after the cat meowed

Matrix vs. Embedded Clauses

Matrix clauses embed embedded clauses.

I heard [that the cat wanted [for there to be cookies]]

root clause | matrix & embedded | embedded

Embedded clause:

① subject/object: NP objects of head V — (I like cats, they said [that we should eat])

② adjunct: optional modifiers — (I saw my cat [after I woke up])

③ relative: CP adjunct of head nouns

I love the cat [that I own]

denote with gap: the cat [that I own __] (object)

My cat ate the pizza [that __ had peppers on it] (subject)

My cat meowed at the dog [who ate his cookies __] (adjunct)

My cat meowed at the dog [that I sent cat food to __] (object of prep.)

Finite vs. non-finite

Finite: inflected for tense or person/number

Non-finite: participles, bare form

Bare form	ING-participle (IP)	P-participle (PP)	Past tense
eat	eating	eaten	ate
walk	walking	walked	walked
sing	singing	sung	sang
be	being	been	was/were
have	having	had	had

Perfect aspect: HAVE + P-participle

present perfect: They have left

past perfect: They had eaten the cake

Progressive aspect: BE + ING-participle

present progressive: He is leaving

past progressive: He was leaving

Bare form: T + bare form

- modal auxiliaries (should, can, ...) * modal is always finite
- to

She should eat.

I want to eat.

Main vs. auxiliary

auxiliary: introduces another non-finite verb, modal or aspectual (other; have, be)

main: doesn't do that

don't mix up main/aux HAVE + BE

I left the house (main)

I have left. (Aux: Asp; P-participle (main))

I must leave (Aux: Mod; following modal is always bare; bare form (main))

I must be eating (Aux: Mod; bare; Aux: Asp; following modal always bare; ING-participle (main))

I am eating (Aux: Asp)

I am happy (main)

I have eaten (Aux: Asp)

I have a cat (main)

Active vs. passive

Active: can have transitive verbs — The cat ate the pizza

Passive: intransitive — The pizza was eaten by the cat

① BE + P-participle

② Demote/remove subject

③ Promote former object to subject

Ex. Lin must (finite modal aux.) have (non-finite bare) slept (non-finite P-participle main)

Lin has (finite aspectual aux.) been (non-finite P-participle asp. aux.) having (non-finite ing-participle main verb) nightmares

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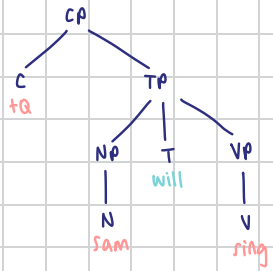
Deep structure (D-structure): output of PSR's

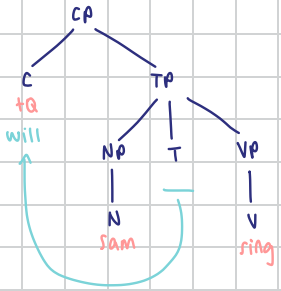
Surface structure (S-structure): output of movement operations

yes/no questions: [+Q]

D-structure

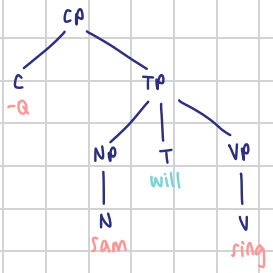


S-structure

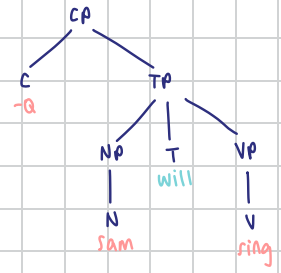


declaratives: [-Q]

D-structure



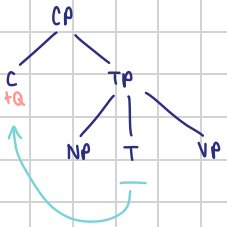
S-structure



T-to-C head movement

A head (T) moves to another head position (C)

Triggered by +Q feature in C, when there is no overt word in C already

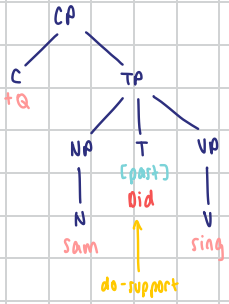


Do-support

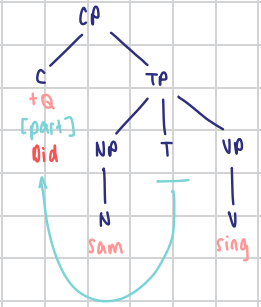
When T contains [past]/[pres], T moves to C [+Q] and "do" is inserted to support both.

Ex. Did Sam sing?

D-structure



S-structure



Aspectual auxiliaries and BE

Finite aspectual auxiliaries and other finite forms of BE can move to C.

Aspectual auxiliaries

Has Sam eaten yet?

Is Sam eating?

Main verb BE

Is Sam at home?

Were the movies bad?

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Phrasal movement

NP movement

The cat ate the cookie (active)

The cookie was eaten (passive)

Wh-movement: moves wh-phrase to left edge of clause

The cat will eat ~~cheddar~~ what cheese.

What cheese will the cat eat? (wh-question)

Wh-movement gaps

Object complement gap

The cat has eaten cheddar cheese.

What cheese has the cat eaten ___?

subject gap

The cat has said I am cool.

Who has the rat said ___ is cool?

PP object complement gap

I don't talk to the cat.

Who don't I talk to ___?

adjunct gap

The cat played soccer in school.

Where did the cat play soccer ___?

Wh-movement in wh-questions

Interrogative wh-NP moves from original pos. in D-structure to CP when C is [+Q].

+Q you have bought what?

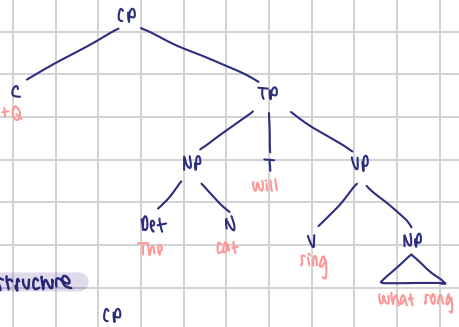
What have you ___ bought ___?

T-to-C

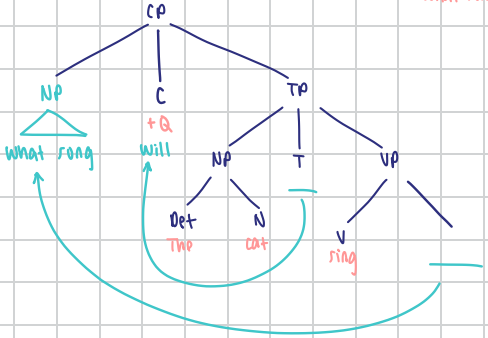
Wh-movement

Ex. What rang will the cat sing?

D-structure



S-structure



Wh-movement: pied-piping

When wh-phrase moves and brings other material

The cat has sent the package to Canada.

The cat has sent the package to which country?

To which country has the cat sent the package ___?

Which country has the cat sent the package to ___?

↑ preposition stranding

How many books did the cat read last month?

Which cookies did the cat like yesterday?

Restrictions on movement

① move does not delete positions in tree

② move can only apply to constituents (phrase/head)

③ superiority: movement only applies to higher candidate

④ islands: some constructions do not allow phrase to move out

coordinate: *Who did the cat see ___ and the student?

The cat saw the dog and the student.

adjunct: *who did the cat call me [CP after he saw ___]?

The cat called me after he saw you

In-situ wh-questions

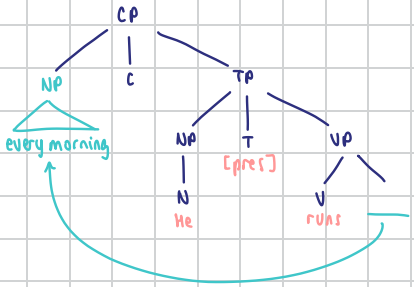
wh-phrase stays in place

The cat put the book on which table?

The cat bought what materials for the project?

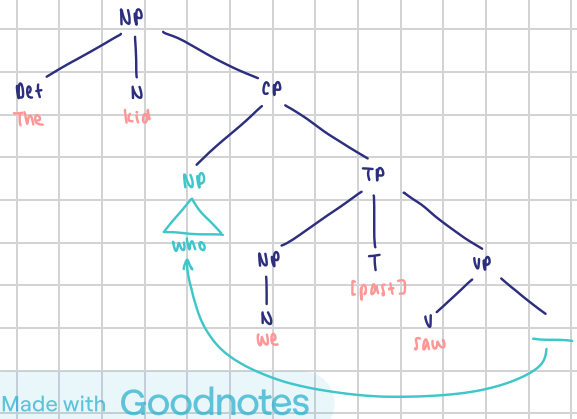
Topicalization

Move phrase to left edge of clause



Relative clauses

The kid [who we saw ___]



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Unit 7: Semantics

Meaning is not specific to language.

Semantics: study of linguistic meaning

Sense & Referent

sense: monarch's representative in Canada

referent: Mary Simon

① **same referent, different sense**

- **referent:** Mark Carney
- **sense:** PM of Canada
- **sense:** leader of Liberal party

② **same sense, different referent**

- **sense:** PM of Canada
- **referent:** Mark Carney, Justin Trudeau

③ **sense, no referent**

- **sense:** present King of France

④ **proper names: only reference**

- **referent:** Mary Simon

sense / referent / denotation

declarative sentences

Ex. "I am sleeping"

- **sense:** truth conditions = true if I am sleeping
- **referent / denotation:** truth / falsehood

you can know sense w/o knowing referent / denotation

> Meaning of sentence = knowing truth conditions

Types of sentences

analytic: always true

"it is what it is", "frozen water is ice"

contradiction: always false

"a duck is not a duck", "cats are fish"

synthetic: truth cannot be determined unless conditions in real world are assessed

"there is a duck on the porch"

Relations between sentences

① **Paraphrase:** same truth conditions

Sam swept the porch <=> the porch was swept by Sam

* rephrasing ≠> paraphrase

② **Entailment:** A => B "A entails B" (assertion)

Sam is a fluffy cat => Sam is a cat

③ **Contradiction:** A => !B negative entailment

Torty is a cat CONTRADICTS Torty is not a cat

④ **Presupposition:** information in sentence assumed to be common ground

Franny's dog is a golden PRESUPPOSES Franny has a dog

Presupposition vs. Entailment

Question-formation cancels entailment but not presupposition

Ex. A. Is Sam a fluffy cat?
B. Sam is a cat
} entailment cancelled (A ≠> B)

Ex. A. Is Franny's dog a golden?
B. Franny has a dog
} presupposition not cancelled

Negation cancels entailment but not presupposition

Ex. A. Sam is not a fluffy cat
B. Sam is a cat
} entailment cancelled (A ≠> B)

Ex. A. Franny's dog is not a golden.
B. Franny has a dog
} presupposition not cancelled

Presupposition triggers

① **Definite descriptions**

↳ definite singular NP's "The president of the university"

↳ possessive NP's "Franny's dog"

② **Factive predicates** e.g. regret

A. Franny regrets that she did not study
B. Franny did not study
} A presupposes B

③ **Aspectual predicates** e.g. stop, start

A. I stopped studying
B. I studied.
} A presupposes B

④ **Temporal clauses** (adjunct clauses)

A. [Before I moved to Tokyo], I ate.
B. I moved to Tokyo
} A presupposes B

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Senses

A lexical item with multiple senses is lexically ambiguous.

① Polysemy: one word w/ multiple senses — same entry in dict

e.g. foot (unit of size), foot (body part)

figurative: a river runs through the town

② Homonymy: different words that sound the same — diff. in dict

e.g. row (pull oars), row (things arranged in a line)

③ Homographs: spelled same (maybe not pronounced same)

e.g. lead (verb), lead (noun)

↳ Heteronym: spelled same, pronounced diff.

e.g. dove (bird), dove (dive past tense)

heteronym ⊂ homograph

Sense relations

① Synonymy: replacing one word doesn't change meaning of sentence — mutually entailing paraphrase

e.g. bucket ≈ pail

"I filled the bucket" = "I filled the pail"

perfect synonymy does not exist (e.g. idioms)

② Antonymy:

↳ complementary antonyms: replacing one word results in contradictory sentences — negation

e.g. "the student is present" / "the student is absent"

↳ gradable antonyms: negation of one is NOT synonymous with the other

e.g. hot/cold, big/small, fast/slow

not big ≠ small

One member of the pair is unmarked, used to ask questions (e.g. how tall are you / # how short are you)

↳ relational opposites: replacing one with other and switching order of arguments results in paraphrases

e.g. give/receive, buy/sell, teacher/student

"I bought a cake from you" = "You sold me a cake"

antonyms must share all but one semantic feature

e.g. handsome / tall — NOT antonyms (dimension of beauty / dimension of height)

handsome / ugly — antonyms

③ Hyponymy: generic-specific relationship

e.g. food / bread

generic (superordinate) — specific (hyponym)

④ Meronymy: part-whole relationship — word naming the part = meronym

e.g. meronyms of body: hand, eye, brain

hyponymy, synonymy, and antonymy are often used in definitions.

Lexical entailment between words means we introduce an entailment relation between sentences that contain there words.

① Synonymy: A lexically entails B and vice-versa (mutual entailment)

A: bucket / B: pail / I filled the bucket <=> I filled the pail

② Antonymy: A lexically entails NOT B and vice versa

A: full / B: empty / This is full => This is NOT empty / This is empty => This is NOT full

③ Hyponymy: B lexically entails A (asymmetric)

A: dog B: golden retriever / This is a golden => This is a dog / This is a dog ≠> This is a golden

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Denotation: what a linguistic expression points to — denoted by ⟦ double brackets ⟧ = "denotation of"

① **proper names:** entities / individuals

⟦Mary Simon⟧ = the individual in the actual world

② **definite descriptions:** entities / individuals

prop noun ⟦the governor general⟧ = the individual in the actual world who is the representative of the Queen in Canada

adj. ⟦orange⟧ = the set of orange entities in the actual world

verb ⟦sneeze⟧ = the set of sneezing individuals in the actual world

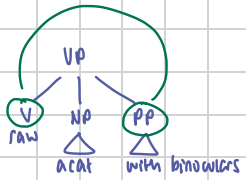
entity ⟦dog⟧ = the set of entities that are dogs in the actual world

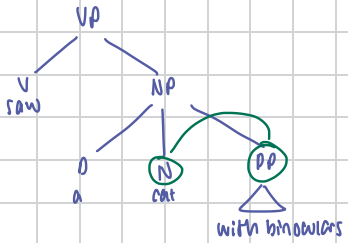
Sense: descriptive content, meaning of a referent mentioned by the expression

Connotation: aspects of meaning not part of sense or denotation. emotional / ideological, conventionalized or personal

Principle of Compositionality: The meaning of a sentence is a function of its parts (words/morphemes) and how they are syntactically arranged.

Modification: modifiers are syntactic siblings of the head they modify





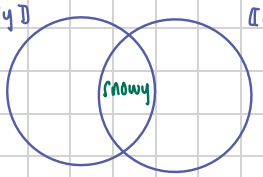
* a modifier has to compose with the head it modifies

Adjectives as modifiers

① **Intersective adjectives** fluffy, red, Canadian, nice

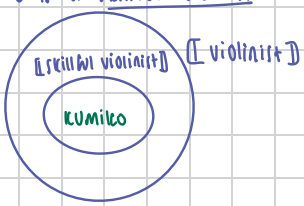
Snowy is a fluffy cat

⟦fluffy⟧ ⟦cat⟧



② **Subsective adjectives** skillful, good, fast

Kumiko is a skillful violinist

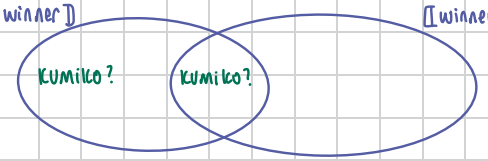


③ **Non-subsective adjectives**

Simple: alleged, possible, 'so-called'

Kumiko is an alleged winner.

⟦alleged winner⟧ ⟦winner⟧



Privative: fake, false, ex-, former

Kumiko is a fake doctor

⟦fake doctor⟧ ⟦doctor⟧

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